Zuoyan Zhang

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Research Interests

Zuoyan Zhang is a Ph.D. student at Hunan University advised by Prof. Jie Zhao, and a visiting Ph.D. student at Instituto Superior Técnico (IST), University of Lisbon, advised by Prof. Nuno Lopes. His research spans AI compilers and large-model training and inference, focusing on automated parallelism, computation and communication overlap, and LLVM optimizations. He also studies floating-point error analysis and polyhedral compilation.

Education

- **Hunan University** *September 2024 - Present*
  Ph.D. in Computer Science and Technology Changsha, China
  - CYCLE Lab at College of Computer Science and Electronic Engineering.
  - Advised by Prof. Jie Zhao.
  - Research on AI compilers and systems for large-model training and inference.
- **University of Lisbon** *August 2026 - Present*
  Visiting Ph.D. Student, Instituto Superior Técnico Lisbon, Portugal
  - Advised by Prof. Nuno Lopes.
  - Research on LLVM optimizations and large-model inference acceleration.
- **Information Engineering University** *September 2021 - June 2024*
  M.Eng. in Computer Technology, School of Cybersecurity Zhengzhou, China
  - State Key Laboratory of Mathematical Engineering and Advanced Computing.
  - Advised by Jinchen Xu and Jie Zhao.
  - Research on floating-point error analysis and polyhedral compilation.
- **Henan University of Technology** *September 2017 - June 2021*
  B.Eng. in Computer Science and Technology Zhengzhou, China
  - College of Information Science and Engineering.

Research & Industry Experience

- **Huawei Technologies Co., Ltd.** *July 2025 - March 2026*
  Compiler Development Intern
  - Implemented automated parallel-configuration search in MindSpeed, reducing search overhead by $5.6\times$ while achieving $1.04\times$ the performance of expert-tuned configurations.
  - Scheduled communication operators at the computation-graph level to overlap communication with computation, achieving $1.14\times$ the performance of SimpleFSDP and $1.10\times$ that of DeepCompile.
  - Developed an agent to derive equivalent expressions for Huawei inference operators, completing support for 20 operators.
- **Beijing Academy of Artificial Intelligence for Science (AISI)** *June 2023 - September 2023*
  High-Performance Computing Intern
  - Enhanced the accuracy-testing framework for core numerical routines in ABACUS.
  - Designed tests for Simpson's integration and real spherical harmonics and integrated them into the testing workflow.

Technical Skills

- Proficient in Python and C/C++, with a strong foundation in object-oriented programming, data structures, and algorithms.
- Distributed large-model training and communication optimization with MindSpeed and Megatron-LM.
- Large-model inference and serving with SGLang and vLLM.
- Deep learning and compiler optimization with PyTorch, MindSpore, and `torch.compile`.
- Familiar with Linux, Git, Docker, Make, and CMake.

Languages

Mandarin (Native) English (Fluent)

Conference Publications

[*] Equal contribution [†] Corresponding author

QRS 2025 **Zuoyan Zhang**, Shihan Yuan, Hongru Yang, Jie Zhao, and Jinchen Xu[†]. 2025. **Scalable Detection of Floating-Point Errors via Adaptive Parallel Subdomain Search**. In *2025 25th International Conference on Software Quality, Reliability and Security (QRS)*. IEEE, 392-403. DOI 10.1109/QRS65678.2025.00047.

ISSTA 2024 Jinchen Xu[*], Mengqi Cui[*], Fei Li, **Zuoyan Zhang**, Hongru Yang, Bei Zhou[†], and Jie Zhao[†]. 2024. **Arfa. An Agile Regime-Based Floating-Point Optimization Approach for Rounding Errors**. In *Proceedings of the 33rd ACM SIGSOFT International Symposium on Software Testing and Analysis* (Vienna, Austria) *(ISSTA '24)*. Association for Computing Machinery, New York, NY, USA, 1516-1528. DOI 10.1145/3650212.3680378.

ASE 2023 **Zuoyan Zhang**, Bei Zhou, Jiangwei Hao, Hongru Yang, Mengqi Cui, Yuchang Zhou, Guanghui Song, Fei Li, Jinchen Xu, and Jie Zhao[†]. 2023. **Eiffel. Inferring Input Ranges of Significant Floating-point Errors via Polynomial Extrapolation**. In *2023 38th IEEE/ACM International Conference on Automated Software Engineering (ASE)*. IEEE, 1441-1453. DOI 10.1109/ASE56229.2023.00139.

Journal Publications

[*] Equal contribution [†] Corresponding author

2026 **Zuoyan Zhang**, Jinyi Wang, Zhuo Tang, and Jie Zhao[†]. 2026. **Global Scheduling of Transient Parameter Materialization for Sharded Training**. *ACM Trans. Archit. Code Optim.* 23, 4 (2026), 1-26. DOI 10.1145/3838601.

2026 **Zuoyan Zhang**, Zhiqiang Zhang, Nanqin Wang, Jinyi Wang, and Jie Zhao[†]. 2026. **Memory-Safe Hierarchical Planning for Hybrid-Parallel Transformer Training**. *ACM Trans. Archit. Code Optim.* 23, 4 (2026), 1-25. DOI 10.1145/3842760.

2026 Shihan Yuan[*], **Zuoyan Zhang**[*], Guanghui Song, Junhui Peng, Feng Wang, Zhuo Tang, Kenli Li, and Jie Zhao[†]. 2026. **A Decoupled Analytical Model for Tile Size Selection in Affine Programs**. *ACM Trans. Archit. Code Optim.* 23, 2, Article 46 (June 2026), 25 pages. DOI 10.1145/3806056.

2024 **Zuoyan Zhang**[*], Jinchen Xu[*], Jiangwei Hao, Yang Qu, Haotian He, and Bei Zhou[†]. 2024. **Hierarchical search algorithm for error detection in floating-point arithmetic expressions**. *J. Supercomput.* 80, 1 (January 2024), 1183-1205. DOI 10.1007/s11227-023-05523-6.

Awards and Scholarships

- **MLIR Summer School Scholarship**, MLIR Foundation *May 2026*
- **CSC Scholarship**, China Scholarship Council *May 2026*
- **PLMW Scholarship**, PLDI 2025 *May 2025*
- **AsiaLLVM Student Scholarship**, LLVM Foundation *April 2025*
- **Outstanding Master's Thesis Award**, Information Engineering University *October 2024*
- **Outstanding Individual in Academic Achievement**, Information Engineering University *June 2024*
- **National Scholarship for Graduate Students**, Ministry of Education *December 2023*
- **First-Class Academic Scholarship**, Information Engineering University *November 2023*
- **Second-Class Academic Scholarship**, Information Engineering University *November 2022*
- **First-Class Academic Scholarship**, Information Engineering University *November 2021*
- **Outstanding Student Leader**, Henan University of Technology *October 2018*

Academic Service

- Artifact Evaluation Committee Member, PACT 2025.
- Teaching Assistant, undergraduate Compiler Principles course, Hunan University.
- Executive Committee Member, CCF Student Chapter at Hunan University.

Selected Projects

- **Intelligent Compiler Optimization of Parallel Strategies for Supernodes** *January 2025 - December 2026*
  Huawei and Hunan University Collaborative Project (TC20241115006) Student Lead
  - Designed hierarchical abstractions of parallel strategies and compiler co-optimization techniques for distributed training of large models.
- **Floating-Point Arithmetic Expression Error Detection** *January 2023 - December 2024*
  State Key Laboratory Open Project (No. 2023B02) Student Lead
  - Developed Maxfpeed, a floating-point error detection tool, and designed algorithms that improved detection performance and effectiveness.
- **Polyhedral-Model-Based Tensor Compiler for Deep Learning** *January 2021 - December 2024*
  NSFC Regional Innovation and Development Joint Fund (No. U20A20226) Contributor
  - Contributed to an automatic mixed-precision code generator that uses a polyhedral model augmented with fitted functions to identify the optimal iteration space for nested-loop programs.
- **Basic Math Library System** *January 2018 - December 2022*
  Subproject of a National Science and Technology Major Project Core Contributor
  - Designed the testing framework for a basic math library on the Sunway supercomputing platform.
  - Wrote automated test code and scripts to improve testing efficiency.

Academic Activities

- **Participant & Scholarship Recipient** *August 2026*
  ACM Europe School on MLIR 2026 A Coruña, Spain
  - Attended courses on MLIR fundamentals and compiler construction.
  - Received a scholarship from the MLIR Foundation to attend the summer school.
- **Presenter** *July 2025*
  QRS 2025 Hangzhou, China
  - Presented SDPS, a scalable approach to detecting floating-point errors in multi-parameter expressions.
  - Explained adaptive domain partitioning and parallel subdomain search for efficient error detection.
- **Participant & Scholarship Recipient** *June 2025*
  PLDI 2025 and Programming Language Mentoring Workshop (PLMW) Seoul, South Korea
  - Attended PLDI 2025 sessions on programming languages and compiler research.
  - Participated in PLMW mentoring activities with scholarship support.
- **Participant & Scholarship Recipient** *June 2025*
  AsiaLLVM Developers' Meeting 2025 Tokyo, Japan
  - Attended technical talks on LLVM, MLIR, and compiler development.
  - Received an LLVM Foundation student scholarship to attend the meeting.
- **Presenter** *September 2024*
  CCF HPC China 2024 Wuhan, China
  - Presented *Optimizing the Computational Performance and Accuracy of Floating-Point Expressions*.
  - Discussed mixed-precision optimization, interval-based expression rewriting, and floating-point error detection.
- **Presenter** *September 2024*
  ISSTA 2024 Vienna, Austria
  - Presented Arfa, a regime-based approach to reducing floating-point rounding errors.
  - Explained how domain partitioning and expression rewriting improve numerical accuracy.
- **Presenter** *September 2023*
  ASE 2023 Luxembourg
  - Presented Eiffel, a tool for inferring input ranges with significant floating-point errors.
  - Explained how clustering and polynomial extrapolation identify error-inducing input ranges.

References

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